

OPERATING SYSTEMS

Storage Management

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File Access Permissions

- Access to files is organized around ownership and protection
- The creator of the file becomes the owner of the file
- Any user will come under a group
- The Access classes for files are
 - **User** – access granted to the owner of file
 - **Group** – access granted to members of same group
 - **Others** – access granted to other normal users

- Commands
 - **chmod** to change the access permission
 - **chown** to change the ownership
 - **chgrp** to change the group owner
- Syntax
 - **chmod** *access-string files*
 - **chown** *new-owner files*
 - **chgrp** *new-group files*

The access permission for the file is

read (symbol r , octal value is 4)

write (symbol w, octal value is 2)

execute (symbol x ,octal value is 1)

Access classes

Users – u

Group – g

Others – o

Access Permission for files can be changed in 2 ways

Relative permissions – symbol representation

Absolute permissions – octal representation

Relative permissions – symbol representation

Adds/removes to the existing

Eg: `chmod ugo+x welcome`

This command provides execute permission for user, group and others on file welcome

+ → adds privilege

- → removes privilege

= → reassigns only specified

Absolute permissions – octal representation

Reassigns the newly specified permissions

Eg: `chmod 444 welcome`

This command changes the file access permission as read for every one on file welcome

Chgrp : Change the group of which the file belongs

Eg: **chgrp** adm welcome

This command changes the file welcome to new group adm

This command needs superuser privilege to execute, Hence add sudo before it

chown : Change the owner of the file

Eg: **chown** root welcome

This command changes file owner as root

This command also needs superuser privilege to execute, Hence add sudo before it



THANK YOU

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